

**A STUDY ON THE INFLUENCE OF SEMESTER
ATTENDANCE ON MARKS OBTAINED BY STUDENTS OF
THE CHEMISTRY AND STATISTICS DEPARTMENTS OF ST.
THOMAS COLLEGE, THRISSUR**

ST. THOMAS COLLEGE (AUTONOMOUS), THRISSUR



PROJECT REPORT

DEPARTMENT OF STATISTICS

**FIRST SEMESTER
M.Sc. STATISTICS
AUDIT COURSE**

P.G. AND RESEARCH DEPARTMENT OF STATISTICS
ST. THOMAS COLLEGE (AUTONOMOUS), THRISSUR
(Affiliated to the University of Calicut)



CERTIFICATE

This is to certify that the project report entitled “**A STUDY ON THE INFLUENCE OF SEMESTER ATTENDANCE ON MARKS OBTAINED BY STUDENTS OF THE CHEMISTRY AND STATISTICS DEPARTMENTS OF ST. THOMAS COLLEGE, THRISSUR**” is a record of Bonafide work carried out by **DEVIKA M, ANANYA C V, and GOURI KRISHNA V N**, and this report is worth submitting for the Degree of Master of Science in Statistics, University of Calicut.

Dr. Anusree M R
Project guide

Place: Thrissur

Date:

DECLARATION

We do hereby declare that this work titled **“A STUDY ON THE INFLUENCE OF SEMESTER ATTENDANCE ON MARKS OBTAINED BY STUDENTS OF THE CHEMISTRY AND STATISTICS DEPARTMENTS OF ST. THOMAS COLLEGE, THRISSUR”** has been originally carried out by us under the guidance of Dr Anusree M R, Assistant Professor, Department of Statistics, St. Thomas College, Thrissur and no part of this study has been submitted or published previously for the award of any other undergraduate degree or post graduate degree.

DEVIKA M (THAZMST007)

ANANYA C V (THAZMST002)

GOURI KRISHNA V N (THAZMST009)

Place: Thrissur

Date:

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DEVIKA M

ANANYA C V

GOURI KRISHNA V N

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Education plays a vital role in the intellectual, social, and professional development of students, and academic performance is often considered a key indicator of educational success. Among the various factors influencing academic achievement, class attendance has consistently been regarded as one of the most significant. Regular attendance enables students to actively participate in classroom discussions, gain a better understanding of course material, and maintain continuous interaction with faculty and peers. Despite its importance, variations in attendance patterns are commonly observed among college students, which may have a direct or indirect impact on their academic outcomes.

In higher education institutions, particularly in science disciplines such as Chemistry and Statistics, classroom learning forms the foundation for conceptual clarity and analytical skill development. These subjects require systematic instruction, problem-solving practice, and continuous engagement, making regular class attendance especially important. St. Thomas College, Thrissur, with its long-standing academic tradition and diverse student population, provides an appropriate setting to examine the relationship between attendance and academic performance.

This project, titled “*A Study On The Influence Of Semester Attendance On Marks Obtained By Students Of The Chemistry And Statistics Departments Of St. Thomas College, Thrissur*” aims to analyse the extent to which class attendance influences students’ academic results. By comparing attendance records with academic performance indicators, the study seeks to identify patterns, measure the strength of association, how alternative study resources influence students to attend lecture classes and highlight the main causes of irregular attendance or missing class hours and factors that make the lecture sections interesting. The findings of this study are expected to contribute to a better understanding of the role of attendance in academic success and may help educators and administrators develop strategies to improve student engagement and performance.

1.2 OBJECTIVES OF THE STUDY

- ☐ To examine students' attendance and marks in the previous semester.
- ☐ To find out whether there is any significant difference in the marks of students having low attendance and high attendance.
- ☐ To find out whether there is a significant association between gender wise marks and attendance.
- ☐ To identify factors that influence students' attendance.

CHAPTER 2

RESEARCH METHODOLOGY

2.1 POPULATION AND SAMPLE

The total number of sampling units in the population is the population size, generally denoted by N, which is 246. In this survey, the population is all the students of the chemistry and statistics departments of St Thomas College, Thrissur.

2.2 SAMPLE SIZE

The sample size is 102 from the chemistry department and 61 from the statistics department. The total sample size is 163.

2.3 SAMPLING TECHNIQUE

The sampling technique selected in administering the questionnaire is stratified sampling. The allocation of sample size in different strata is done by proportional allocation. Samples are collected from each stratum by simple random sampling without replacement.

2.4 DETERMINATION OF SAMPLE SIZE

In order to determine the sample size for the main survey, we have used the following equation;

$$n = (((Z^2 p(1 - p))/E^2))/(1 + (Z^2 p(1 - p))/E^2)$$

Where,

n = required sample size

N = population size

Z = Z-score corresponding to the confidence level (1.96 for 95% CI)

p = estimated proportion of the population (if unknown, use 0.5 for maximum variability)

E = margin of error (e.g., 0.05 for a 5% margin of error)

By using the above equation, we have found out that our sample size is 102 from the chemistry department and 61 from the statistics department. The total sample size is 163.

2.5 SOURCE OF DATA

A set of questions is prepared with Google Forms according to the objectives, and it is distributed among a sample of the population through WhatsApp groups or as personal messages. Hence questionnaire method is used for the collection of data. 17 questions were required to collect data for the evaluation of the objectives. The questionnaire was prepared by the team members.

2.6 STATISTICAL CHARTS USED

- ❖ Bar Chart
- ❖ Pie Chart

2.7 STATISTICAL TESTS USED

- ❖ Chi-Square Test for Independence
- ❖ Mann-Whitney U Test
- ❖ Spearman's Rank Correlation

CHAPTER 3

STATISTICAL ANALYSIS

3.1 A. GENDER DISTRIBUTION

The pie chart showing the percentage of male and female students who attended the survey from the Statistics Department is given below:

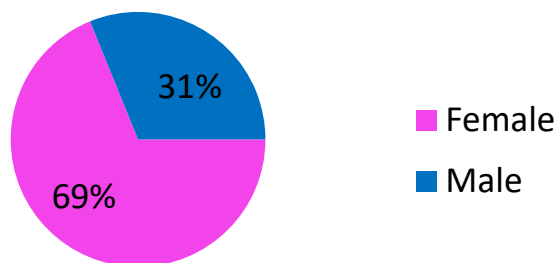


Figure 3.1.1 The pie chart showing the percentage of male and female students who attended the survey from the Statistics Department

From the above pie chart, it is observed that 69% of the respondents are Females, while 31% are Males. This shows that female students have a higher participation rate in data submission from this department.

3.1 B. GENDER DISTRIBUTION

The pie chart showing the percentage of male and female students who attended the survey from the Chemistry Department is given below:

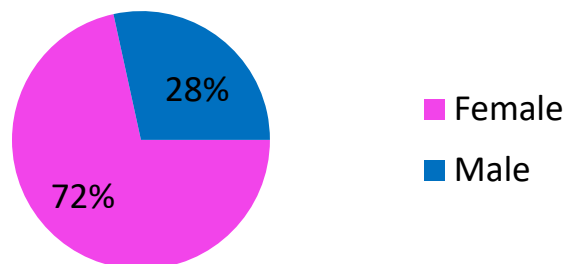


Figure 3.1.2 The pie chart showing the percentage of male and female students who attended the survey from the Chemistry Department

From the above pie chart, it is observed that 72% of the respondents are Females, while 28% are Males. This shows that female students have a higher participation rate in data submission from this department.

3.2 MARK FREQUENCY

Here we are observing how the number of students varies with different mark intervals. The bar chart and frequency table that display the distribution of students' marks in percentages are given below:

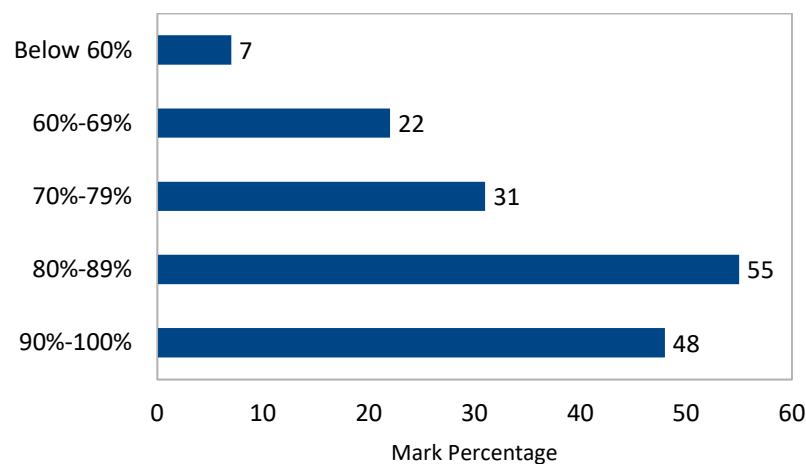


Figure 3.2.1: Bar chart of frequency of mark distribution of students

Mark percentage	Frequency	Percentage
90%-100%	48	29.00%
80%-89%	55	34.00%
70%-79%	31	19.00%
60%-69%	22	13.00%
Below 60%	7	4.00%
Total	163	100.00%

Table 3.2.1: Frequency of mark distribution of students

The above table shows that the majority of students (55 out of 163) fall in the 80 -89% mark, making it the most frequent interval. Only 7 students (4%) scored below 60%, showing minimal underperformance.

3.3 ATTENDANCE DISTRIBUTION

The Pie chart and Frequency table display the distribution of students' attendance in percentages.

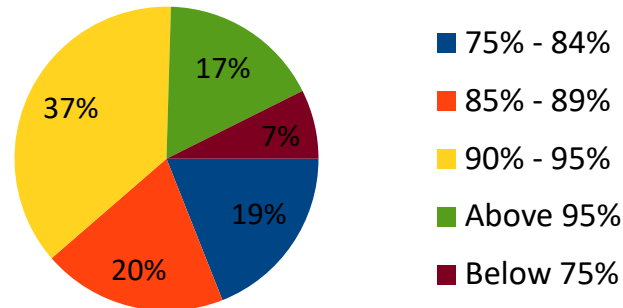


Fig 3.3.1 Pie chart of attendance distribution of students

Attendance	Frequency	percentage
Below 75%	12	7.00%
75% - 84%	31	19.00%
85% - 89%	32	20.00%
90% - 95%	60	37.00%
Above 95%	28	17.00%
Total	163	100.00%

Table 3.3.1: Frequency table of attendance distribution of students

Most students (37%) have 90 -95% attendance, indicating a generally good attendance pattern, while 7% fall below 75%, signalling potential concern.

3.4 FREQUENCY OF CLASS BUNKING

Here we are observing how often classes are bunked by students. The Bar chart and Frequency table that display how often classes are bunked is given below:

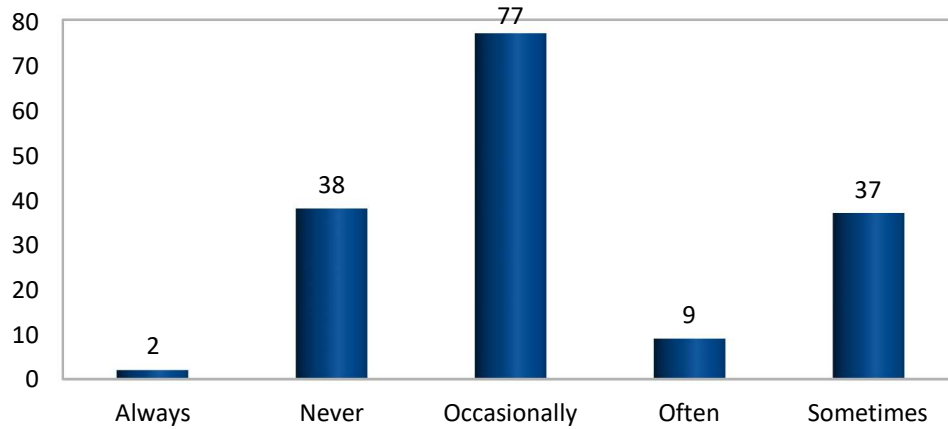


Figure 3.4.1 Bar chart of the frequency of how often classes are bunked

How often are classes bunked	Frequency	Percentage
Always	2	1.2%
Never	38	23.3%
Occasionally	77	47.2%
Often	9	5.5%
Sometimes	37	22.7%
Total Result	163	100%

Table 3.4.1: Frequency table of class bunking

The majority of respondents skip classes only occasionally (47.2%), while nearly a quarter never skip them. A small fraction (1.2%) always bunk classes.

3.5 REASONS FOR IRREGULAR ATTENDANCE

Here we are observing the reasons for irregular attendance. The Bar chart displays the causes of irregular attendance is given below:

Causes of Irregular Attendance

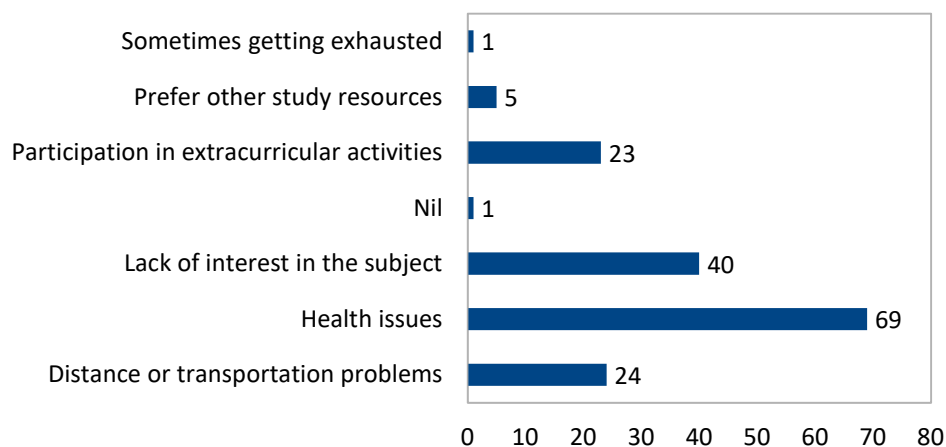


Fig 3.5.1: Bar chart of the frequency of the cause of irregular attendance

Health issues are the dominant factor, with the largest number of instances (~70). This suggests that students miss classes mainly because of medical problems. The data highlights that improving health support and addressing transportation challenges would likely reduce irregular attendance.

3.6 ATTENDANCE DISTRIBUTION OF CHEMISTRY AND STATISTICS DEPARTMENTS

The bar chart illustrates the number of students in the chemistry and statistics departments categorized by their attendance percentage.

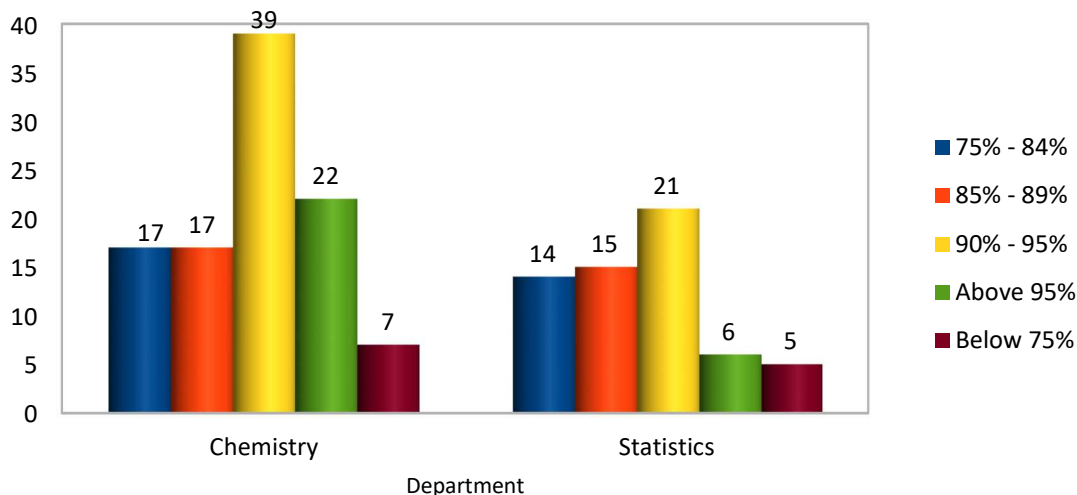


Fig 3.6.1: Bar chart of attendance distribution of chemistry and statistics departments

The above figure shows that the Chemistry department has a higher percentage of students with attendance of 90-95% (38%), indicating better overall attendance. The Statistics department shows a concentration of students in the 90–95% range (34%). Both departments have similar proportions in the below 75% and 75–84% ranges.

3.7 MARKS DISTRIBUTION BASED ON GENDER

The bar chart showing the distribution of female and male students across different marks is given below:

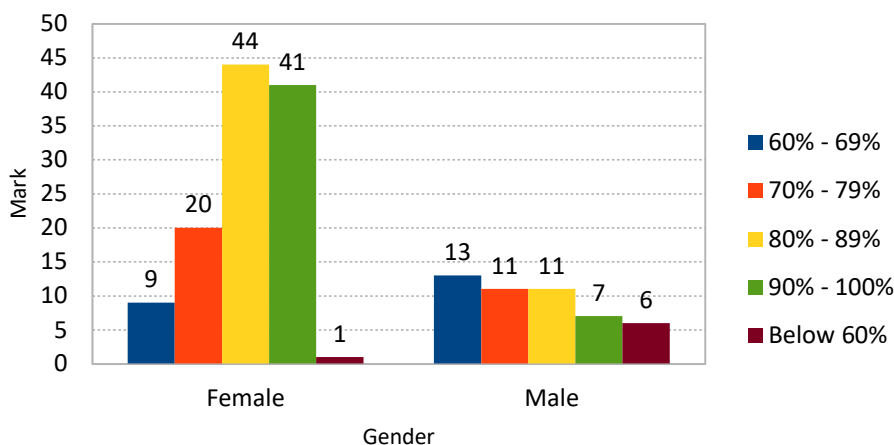


Fig.3.7.1: Bar chart of marks distribution based on gender

The above figure shows that Female students outperform male students, especially in the higher mark ranges (80% and above). Male students show greater variability, with a noticeable proportion scoring below 60%. Academic performance appears stronger and more consistent among females.

3.8 ATTENDANCE DISTRIBUTION BASED ON GENDER

The bar chart showing the distribution of female and male students across different attendance ranges is given below:

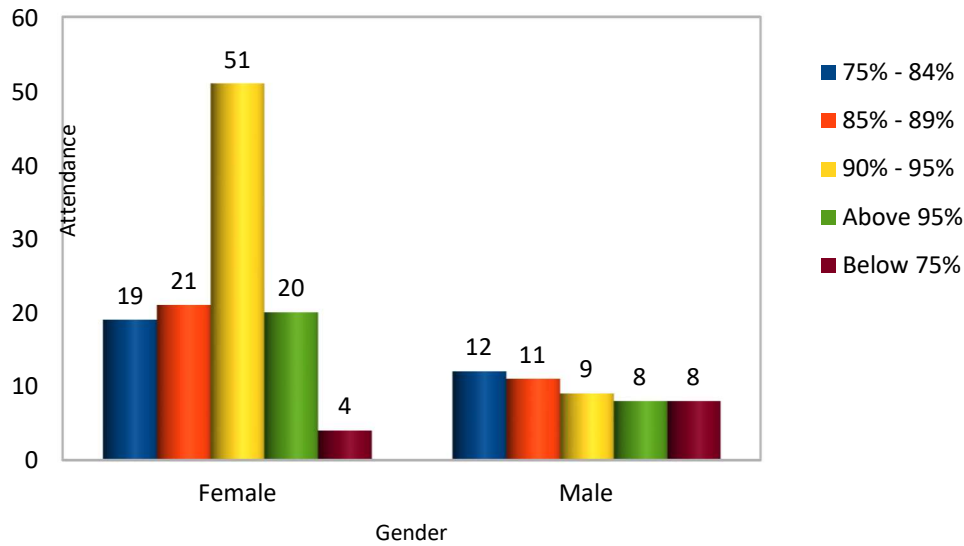


Fig 3.8.1: Bar chart of attendance distribution based on gender

The chart reveals that female students exhibit more consistent and higher attendance levels than male students, while male students show greater variability and comparatively lower attendance in some categories.

3.9 FACTORS AFFECTING LIVE LECTURE INTEREST AND ATTENDANCE

The bar chart showing the factors affecting live lecture interest and attendance is given below:

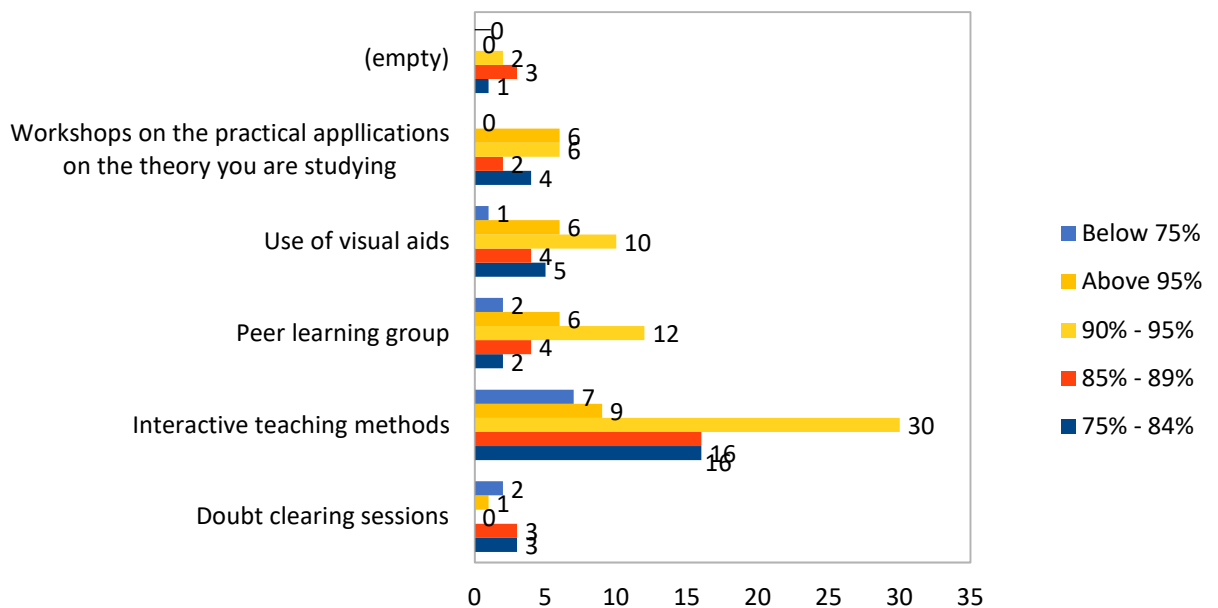


Fig 3.9.1: Bar chart of factors affecting live lecture interest and attendance

The graph shows that Peer learning groups have the strongest positive influence on lecture interest and attendance. Visual aids and interactive teaching also boost attendance. Workshops have a moderate effect, while doubt-clearing sessions show no measurable impact in the chart.

3.10. INFLUENCE OF ALTERNATIVE STUDY RESOURCES ON LIVE LECTURE BUNKING

We are observing the responses of students regarding how alternative study resources affect their decision to bunk live lectures. The pie chart showing the responses is given below:

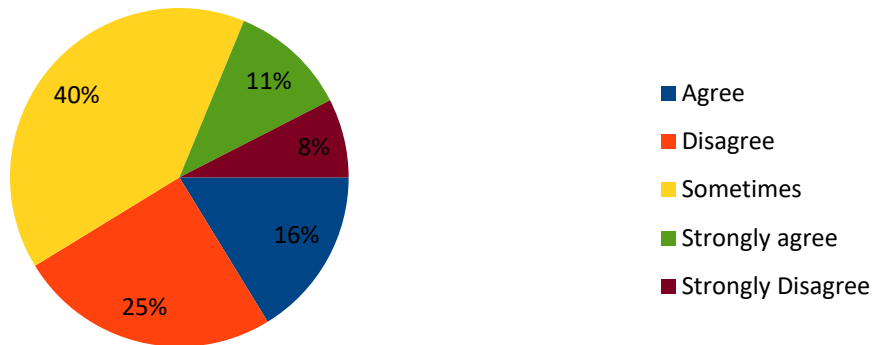


Fig 3.10.1: Pie chart of responses

The graph shows that the largest segment (40%) chose Sometimes, indicating that alternative resources occasionally influence lecture bunking. Combined disagree responses (25% + 8% = 33%) indicate a larger group believes the alternative study resources do not influence lecture bunking.

3.11 INFLUENCE OF ATTENDANCE ON MARKS

The bar chart shows the percentage of students with different attendance levels and their corresponding mark distributions given below:

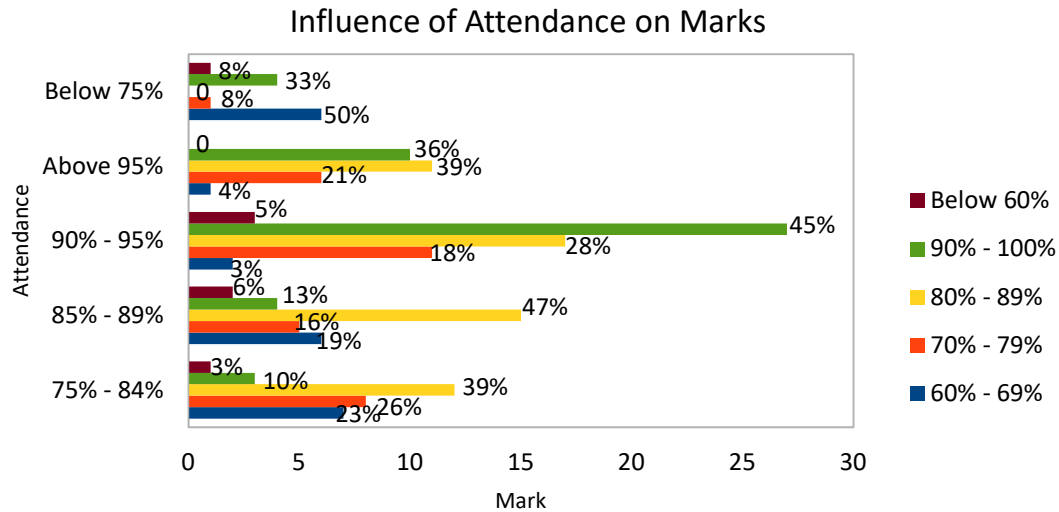


Fig 3.11.1 Bar chart of the influence of attendance on marks

The graph shows that students with 85–89% attendance have the highest proportion (47%) scoring 80-89% marks. Those with below 75% attendance show a mixed distribution, with 33% in 90–100% and 8% in Below 60%. Generally, higher attendance correlates with better mark performance, but the “Above 95%” group shows a slightly lower top-score percentage than the 90–95% group.

3.12: ASSOCIATION BETWEEN GENDER AND ATTENDANCE

In this analysis, we will examine the relationship between gender and mark.

H0: Gender and attendance are independent

H1: Gender and attendance are dependent

		2		Total
		Male	Female	
Below 75%	Count	8	4	12
	Expected Count	3	8	12
75% - 84%	Count	12	19	31
	Expected Count	9	22	31
85% - 89%	Count	11	21	32
	Expected Count	9	23	32
90% - 95%	Count	9	51	60
	Expected Count	18	42	60
Above 95%	Count	8	20	28
	Expected Count	8	20	28
Total	Count	48	115	163
	Expected Count	48	115	163.

Table 3.12.1: chi square contingency table of gender and attendance

	Value	df	Asymp. Sig. (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)	Point Probability
Pearson Chi-Square	15.693 ^a	4	.003	.003		
Likelihood Ratio	15.524	4	.004	.005		
Fisher's Exact Test	15.422			.003		
Linear-by-Linear Association	9.213 ^b	1	.002	.003	.002	.001
N of Valid Cases	163					
Cramer's V	.310					

Table 3.12.2:Result of Chi-Square Test

Since the p-value is less than 0.05, there is a significant association between gender and attendance, ie, gender is a factor that influences attendance. Since the Cramer's V value is .310, there is a moderate association between gender and attendance.

3.13 ASSOCIATION BETWEEN GENDER AND MARK

In this analysis we will examine the relationship between gender and mark.

H0: Gender and mark are independent

H1: Gender and mark are dependent

		2		Total
		Male	Female	
Below 85%	Count	7	41	48
	Expected Count	14	34	48
85% - 89%	Count	11	44	55
	Expected Count	16	39	55
90% - 95%	Count	11	20	31
	Expected Count	9	22	31
Above 95%	Count	19	10	29
	Expected Count	8	20	29
Total	Count	48	115	163
	Expected Count	48	115	163

Table 3.13.1: pooled chi square contingency table of gender and mark

	Value	df	Asymp. Sig. (2-sided)	Exact Sig. (2-sided)	Exact Sig. (1-sided)	Point Probability
Pearson Chi-Square	26.171 ^a	3	.000	.000		
Likelihood Ratio	24.982	3	.000	.000		
Fisher's Exact Test	24.464			.000		
Linear-by-Linear Association	23.197 ^b	1	.000	.000	.001	.000
N of Valid Cases	163					
Cramer's V	.401					

Table 3.13.2: Result of Chi-Square Test

Since the p-value is less than 0.01, there is a significant association between gender and mark, ie, gender is a factor that influences the mark. Since the Cramer's V value is .410, there is a strong association between gender and attendance.

3.14 RELATIONSHIP BETWEEN MARK AND ATTENDANCE

In this analysis, we will examine the association between mark percentage and attendance percentage.

H0: There is no significant relationship between mark percentage and attendance percentage

H1: There is significant relationship between mark percentage and attendance percentage

Spearman's rank correlation rho	
S = 495683	p-value = 4.676e-05

Table 3.14.1: Result of Spearman's correlation

p-value is less than 0.05, so the association is statistically significant, and attendance is associated with marks.

3.15 MARKS COMPARISON ON HIGHER AND LOWER ATTENDANCE PERCENTAGE

We get from the previous result that attendance and marks are correlated. To make it more reliable, we are testing whether there is any significant difference in the marks scored by the high attendance group and the low attendance group. For that, we are using Mann-Whitney U test.

H0: There is no significant difference in the marks scored by the high attendance group and the low attendance group.

H1: There is a significant difference in the marks scored by the high attendance group and the low attendance group.

Wilcoxon rank sum test with continuity correction	
W	17724
p-value	1.408e-07

Table 3.15.1: Result of Mann-Whitney U test

p-value is less than 0.05, so there is a significant difference in the marks between high attendance and low attendance groups.

CHAPTER 4

SUMMARY AND CONCLUSION

4.1 SUMMARY

This study was conducted to analyze the influence of class attendance on the academic performance of students in the Chemistry and Statistics departments of St. Thomas College, Thrissur. The study also examined the role of gender in influencing attendance and academic achievement. Statistical analyses were carried out to understand the relationships among gender, attendance, and marks obtained by students. The results of the study revealed a significant association between gender and attendance, indicating that gender is an important factor influencing students' attendance patterns. This suggests that male and female students differ significantly in their level of participation in regular classroom activities. Further, the study found a significant association between gender and academic performance, which shows that gender also has an influence on students' marks. This implies that academic achievement varies significantly between male and female students in the selected departments. Most importantly, the analysis of the relationship between attendance percentage and mark percentage showed a significant difference between the academic performance of students with high attendance and those with low attendance. Students who maintained higher levels of class attendance performed significantly better academically compared to students with lower attendance. This clearly demonstrates that regular class attendance has a positive impact on academic performance.

4.2 CONCLUSION

- The analysis shows that the majority of students maintained high attendance levels, with most students falling in the 90%–95% attendance category. Only a small proportion of students had attendance below 75%, indicating that irregular attendance is limited to a minority. Department-wise comparison reveals that Chemistry students exhibit slightly better attendance than Statistics students, though both departments show satisfactory attendance patterns.
- Students with higher attendance percentages consistently achieve higher marks, particularly in the 80% and above mark categories. Lower attendance levels are associated with lower academic performance, with a noticeable concentration of low scores among students with attendance below 75%. The graphical representation clearly demonstrates a positive relationship between attendance and academic achievement, emphasizing the role of regular class participation in academic success. A clear difference is observed between the academic performance of students with high attendance and those with low attendance.
- Gender-wise analysis shows that female students outperform male students academically, especially in higher mark ranges. Female students also exhibit better attendance patterns, while male students show greater variability in both attendance and marks. The association between gender and attendance, as well as gender and academic performance, indicates that gender plays a role in influencing both attendance behaviour and academic outcomes. Overall academic performance is found to be more consistent among female students than male students.
- Health-related issues are identified as the primary cause of irregular attendance, followed by transportation difficulties. Participation in extracurricular activities and lack of interest in certain subjects also contribute to occasional absenteeism. Peer learning groups, interactive teaching methods, and the use of visual aids are found to positively influence students' interest in live lectures and attendance. Alternative study resources, such as

recorded lectures and online materials, influence class skipping only occasionally, suggesting that they do not fully replace classroom learning

- Most students acknowledge that regular class attendance positively affects academic performance. Students recognize that classroom interaction, peer discussion, and direct engagement with teachers contribute to better understanding of subject content. The overall perception of students supports the idea that attendance is an essential component of academic success, rather than merely a formal requirement.
- The study shows that class attendance has a strong and positive influence on academic performance among students of the Chemistry and Statistics departments. Higher attendance is consistently associated with better academic outcomes, while irregular attendance is linked to lower performance. Factors such as health issues, transportation problems, and teaching methodologies significantly affect attendance patterns. Enhancing classroom engagement, improving student support systems, and encouraging regular attendance can contribute significantly to improving students' academic achievement.

4.3 SCOPE OF THE STUDY

1. The study helps teachers and college administrators understand how class attendance affects the academic performance of students in the Chemistry and Statistics departments. By identifying the extent of this impact, educators can plan better teaching strategies, introduce remedial classes for irregular students, and strengthen attendance monitoring systems to improve academic results in the future.
2. The findings of this study make students aware of the importance of regular class attendance for achieving better academic performance. It encourages students to develop discipline, improve learning consistency, and take responsibility for their academic progress, which can positively influence their future educational and career outcomes.

3. The study provides evidence-based information that can be used by college authorities while framing or modifying attendance policies, academic support programs, and student counselling services. Such policies can help improve overall academic standards and reduce poor performance caused by irregular attendance.

4. This study serves as a foundation for future researchers who wish to conduct similar studies in other departments, colleges, or universities. Researchers can extend the scope by including more subjects, larger samples, or additional variables such as teaching methods, motivation, or socio-economic factors, thereby contributing to further academic research in the field of education.

4.4 LIMITATIONS OF THE STUDY

1. **Limited Sample Size:** The study is confined to students of the Chemistry and Statistics Departments of St. Thomas College, Thrissur. Therefore, the findings may not be fully representative of students from other departments or institutions.
2. **Single Institution Focus:** Since the research is conducted only in St. Thomas College, Thrissur, the results cannot be generalized to other colleges or universities with different academic environments and academic policies.
3. **Limited Variables Considered:** The study focuses mainly on class attendance as a factor influencing academic performance. Other important factors, such as study habits, teaching methods, motivation levels, family background, and socio-economic conditions, are not considered.
4. **Dependence on Self-Reported Data:** The study is based on data collected directly from students through questionnaires or interviews. As a result, the accuracy of the findings depends on the honesty, memory, and perception of the respondents, which may lead to response bias or inaccuracies.

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ANNEXURE

1) Gender:

- a) Male
- b) Female

2) Current level of study:

- a) UG
- b) PG

3) Course/Program Name:

- a) Statistics
- b) Chemistry

4) Current year:

- a) 1st
- b) 2nd
- c) 3rd

5) Please indicate your average attendance percentage for the previous semester:

- a) Below 75%
- b) 75% - 84%
- c) 85% - 89%
- d) 90% - 95%
- e) Above 95%

6) Please select your mark percentage range from the previous semester:

- a) 90% - 100%
- b) 80% - 89%
- c) 70% - 79%
- d) 60% - 69%
- e) Below 60%

7) Do you think classroom hours help you achieve your current percentage of marks?

- a) Yes
- b) No

8) Do you use YouTube or other online resources in addition to regular class lectures?

a) Yes

b) No

9) How effective are these alternative resources in helping you score well in semester exams?

a) Very effective

b) Somewhat effective

c) Not effective

d) Not used

10) Alternative study resources influence you to skip live lectures

a) Strongly agree

b) Agree

c) Sometimes

d) Disagree

e) Strongly Disagree

f) Other:

11) Do you think attendance plays a major role in the marks you scored?

a) Yes

b) No

12) Do you think the marks you scored as low?

a) Yes

b) No

13) If yes what may be the reasons?

a) Low attendance

b) Lack of regular study

c) Difficulty in understanding subjects

d) Exam stress/anxiety

e) Distractions from social media

f) Other:

14) Do you think you have a low attendance percentage?

a) Yes

b) No

15) How often do you skip classes?

a) Never

b) Occasionally

c) Sometimes

d) Often

e) Always

16) What are the main causes of irregular attendance or missing class hours?

a) Health issues

b) Lack of interest in the subject

c) Participation in extracurricular activities

d) Distance or transportation problems

e) Prefer other study resources

f) Other:

17) Which factors make live lectures interesting and motivate you to attend them?

a) Interactive teaching methods

b) Use of visual aids

c) Peer learning group

d) Workshops on the practical applications of the theory you are studying

e) Doubt-clearing sessions

[Google form](#)